ScAN - A Path to PIM Hardware-aware Network

Phase Roadmap

Phase 1 Float Training AllCNN-C CIFAR-100 FP32	**Phase 2** W4A4 QAT LUTQ / uniform W4 PACT activations integer simulator	**Phase 3** KD Finetune EfficientNet teacher SS / CS / TU	**Phase 4** PIM Row Sparsity 2/3 rows off ADC7 + digital BN current tuning
Phase 5 PCN Integration 3-iter fb/ff refine sensitive channels beta=1/8	**Phase 6** HW Noise Finetune real F/G/H LUT voltage readout noise model	**Final Eval** HW simulator full noise model multi-seed	

Phase Results

Phase	Experiment	Precision / path	Acc.	Notes
1	RGB AllCNN-C + BN	FP32 float	72.99%	Standard CIFAR-100 RGB, 3x32x32, crop/flip augmentation; checkpoints/ALL_CNN_C_c100_rgb_aug_bn_refstyle.pth.
1	RAW/RGGB AllCNN-C	FP32 float	61.31%	Current 4-channel RGGB BN baseline, raw_noise=True.
2	RAW/RGGB W4A4	uniform W4A4 INT sim	56.61%	Validated noisy simulator baseline.
2	RAW/RGGB W4A4	uniform W4A4 INT sim	57.09%	Clean simulator reference.
2	non-uniform SS	integer LUT-Q m=7	56.78%	Integer LUT values [-7,...,7].
2	non-uniform SS	integer LUT-Q m=15	56.52%	Integer LUT values [-14,-12,...,14].
2	non-uniform SS + PACT	I16/W4/A4 LUT-Q m=15	60.21%	Fixed integer simulator, raw_noise=True.
3	QKD CS + PACT	I16/W4/A4 LUT-Q m=15	61.24%	Co-studying result from simulator.
3	QKD TU + PACT	I16/W4/A4 LUT-Q m=15	61.66%	Current best dense int4 simulator result.
4	PIM-QAT BN-only	spec W4/A4/ADC7	49.63%	First completed PIM row-sparse run; target is 57-58, needs tuning.

Teacher / KD

Model / stage	Acc.	Notes
EfficientNetV2-L teacher, clean	77.30%	Verified on current H5 test split.
EfficientNetV2-L teacher, noisy	77.33%	Default noisy evaluation.
Co-studied teacher, noisy	78.70%	Earlier CS run improved teacher.
QKD SS rerun	60.21%	I16/W4/A4 LUT-Q + PACT, fixed simulator.
QKD CS rerun	61.24%	I16/W4/A4 LUT-Q + PACT, fixed simulator.
QKD TU rerun	61.66%	Best current dense int4 simulator result.

Current PIM Setting

Item	Current setting
Model	ALL-CNN-C, 4-channel RGGB input, BatchNorm; GBN and PCN disabled for the minimal pass.
Weights	LUT-Q 4-bit with deployed integer codebook [-7,...,7] and precomputed per-8-output-channel scales.
Activations	First layer input is unsigned I16; hidden activations are unsigned A4 codes in [0,15] and values in [0,1].
Rows	Stage1.0 and logits are unmasked. Masked layers keep 48/144 rows; stage1.1 hybrid keeps 72/144 rows.
ADC	7-bit mid-rise tile ADC, fixed window W=K, no read noise in the current training run.
Simulator	PIM-QAT forward and PIM simulator are numerically matched on smoke tests; deployed inference uses integer weights/activations and precomputed constants.

Next

Improve Phase 4 PIM-QAT accuracy from the current 49.63% toward 57-58%. Add pim_info/harware_model.tex voltage-domain readout as an optional hardware simulator once F/G/H and noise parameters are specified. Add PCN only after the minimal BN row-sparse PIM path is stable.